Nabeel Ahmed Kabir

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Education

York University **Expected Graduation: April 2028**
Bachelor of Computer Engineering *Toronto*

- Courses: Data Structures and Algorithms, Design and Analysis of Algorithms, Software Design, Operating System Fundamentals, Computer Networks / Network Protocols, Computer Organization, Digital Logic Design

Work Experience

Cotriply **Apr 2026 – Jun 2026**
AI Software Developer Intern *Toronto, Canada*

- Transitioning an **AI-powered group travel coordination platform** from prototype to MVP as part of the **Advance Ontario** program
- Refactored **16+ monolithic files** (some **6,000+ lines**) into modular **TypeScript** components, reducing codebase size by **50%** with zero errors
- Built an automated **email notification system** delivering **real-time alerts** across all core trip events (document uploads, polls, itinerary updates), dynamically scaling to notify every member regardless of trip size
- Configured a **Vercel** staging environment integrated with **GitHub**, decoupling development from the live platform
- Resolved **10+ bugs** across trip creation, UI overflow, poll logging, and itinerary auto-save via **Postman** API testing

Projects

Distributed Task Queue | *Demo* | *Source Code* **Go | Redis | PostgreSQL | Docker**

- Built a **fault-tolerant distributed task queue** in **Go** using **Redis Streams** for at-least-once delivery, with atomic job claiming, **crash recovery (XAUTOCLAIM)**, retry with exponential backoff, dead-lettering, and idempotency checks
- Verified **priority-based scheduling** through integration testing against real **Dockerized** Redis/Postgres/worker processes (no mocks) – high-priority jobs submitted last were claimed **2x faster** than low-priority jobs submitted first (4.6s vs 11.7s avg wait)

Comms-Denied Autonomy Hackathon | *3rd Place, $1000 – Red Team DefTech* **Python | OpenCV**

- Built a fully autonomous **drone navigation system** in **Python** using **OpenCV** to interpret visual cues for real-time flight decisions with **zero human intervention or GPS**
- Designed a **computer vision pipeline** using **HSV color masking** and **contour analysis** to detect and classify objects, replacing an unreliable LLM-based approach for improved speed and reliability
- Collaborated with a **5-person team** under a **9-hour constraint**, owning the vision pipeline and decision-logic components; placed **3rd of 10 teams**

FPGA Snake Game | *Demo* | *Source Code* **Verilog | Quartus Prime | DE10-Lite**

- Implemented **Snake** entirely in hardware on an **FPGA** using a multi-state **Finite State Machine (FSM)** in **Verilog**, with **VGA signal generation** and clock division driving real-time rendering
- Built collision detection (walls and self) and score tracking logic, collaborating with a partner to resolve hardware input reliability issues through debouncing and timing control

Evolutionary AI Racer | *Source Code* **Python | Pygame | Neural Networks**

- Built an evolutionary AI racing simulator in **Python** and **Pygame** where **neural-network**-controlled agents learn steering and movement through a **genetic algorithm** (selection, crossover, mutation)
- Improved lap completion rate from near **0% to 70%+** over **300 generations**, reducing iteration time by **50%** through modular, replay-driven debugging

Technical Skills

Languages: Python, JavaScript, TypeScript, Go, Java, C, SQL, HTML5/CSS3, Bash, Verilog

Developer Tools: Git, GitHub, VS Code, Docker, Postman, Microsoft Azure, Vercel, Linux/Unix, Spring Boot, Claude Code, Quartus Prime

Libraries/Frameworks: React, OpenCV, pandas, NumPy, scikit-learn, Leaflet.js, Redis, MySQL, PostgreSQL, MongoDB

Core Competencies: OOP, REST APIs, JUnit, CI/CD, Agile/Scrum, Prompt Engineering, LLM Integration, Agentic Workflows